

Vamsi Vajja

Senior AI/ML Engineer

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🌐 Portfolio

Professional Summary

I am a **Senior AI/ML Engineer** with over 8+ years of experience designing and delivering scalable, production grade data and machine learning systems across domains such as e-commerce, mobility, retail, and financial services. I have developed enterprise grade AI systems using **Retrieval Augmented Generation (RAG)**, **Large Language Models (LLMs)**, **LangChain** and **vector databases** like **FAISS** and **OpenSearch**, enabling context aware, compliant knowledge assistants. I have led the development of intelligent platforms that transform large scale, real time data into actionable insights and automated decision making systems. My expertise spans the full ML lifecycle from data ingestion and feature engineering to model training, deployment, and monitoring leveraging **Python**, **SQL**, **PyTorch**, **TensorFlow**, and **Scikit learn**. I have built robust data pipelines and real-time processing systems using **Apache Kafka**, **Celerys** and **Redis**, ensuring high throughput and low latency performance in distributed environments. My work includes building validation engines with rule based systems, implementing pricing and content guardrails, and ensuring data quality at scale using optimized queries in MySQL and caching strategies like Memcached. From a systems perspective, I specialize in designing cloud native and microservices based architectures using **AWS (Lambda, S3, CloudWatch, API Gateway)**, **Docker**, **Kubernetes**, and **FastAPI/Node.js** for scalable backend services. I have implemented MLOps pipelines using **MLflow**, **CI/CD tools** like **Jenkins** and **Azure DevOps**, and ensured observability through logging and monitoring frameworks. My ability to bridge data science and engineering allows me to build highly reliable, efficient, and production ready AI systems.

Skills

GenAI & LLMs	Agentic AI, RAG, LLM-as-a-Judge, Fine-tuning, Prompt Engineering, LangChain, LlamaIndex, vLLM, DeepSeek, OpenAI API
Machine Learning	PyTorch, TensorFlow, Hugging Face, Scikit-learn, XGBoost, LightGBM, Sentence Transformers, Deep Learning
Data Platforms	Databricks (Delta Lake, Unity Catalog), Spark, PySpark, Kafka, Airflow, Flink, Exactly Once Semantics
Cloud & Infrastructure	AWS Bedrock, GCP Vertex AI and BigQuery, Azure OpenAI, Cognitive Search, Docker, Kubernetes, Terraform
MLOps & Monitoring	MLflow, DVC, LangSmith, Weights & Biases, Arize AI, Fiddler AI, Prometheus, Grafana, GitHub Actions
Databases	Vector DBs (OpenSearch, FAISS, Chroma), PostgreSQL, MongoDB, DynamoDB, Redis Streams
Languages	Python, SQL, C, Java, Bash, JavaScript
API & Security	FastAPI, Flask, OAuth2, JWT, PII Redaction, MCP, FastMCP

Work Experience

Barclays Whippany, NJ <i>Senior AI/ML Engineer</i>	Jan 2024 – Present
Overview: Barclays US Consumer Bank identified manual After Call Work (ACW) as a key bottleneck in contact center operations, where agents spent significant time summarizing every customer interaction post call. This project leveraged Generative AI to automatically capture call intent, agent actions, compliance flags, and next steps from raw transcripts, eliminating manual effort and enabling agents to focus entirely on customer experience. Developed an enterprise grade GenAI knowledge assistant using LangChain and LangGraph, enabling users to query financial policies and compliance documents using natural language. Designed the system for scalable deployment on AWS to support high performance enterprise usage.	

- Implemented a **Retrieval Augmented Generation (RAG)** pipeline to ground LLM responses in enterprise knowledge sources, integrating with **Amazon Bedrock** for foundation models. Ensured context aware, reliable outputs while minimizing hallucinations in domain-specific queries.
- Deployed a vector based semantic search system by generating embeddings and indexing them in **Amazon OpenSearch and FAISS**, enabling efficient similarity search with sub second (<500ms) retrieval latency.
- Built scalable document ingestion and preprocessing pipelines to extract and process unstructured data from PDFs and internal repositories, storing raw documents in **Amazon S3**. Applied intelligent chunking strategies to preserve semantic context and improve retrieval quality.
- Implemented advanced document chunking techniques using **LangChain** text splitters, including recursive and fixed size **chunking** with overlap. Optimized chunking strategies to improve contextual grounding and increase retrieval relevance by **30-35%**.
- Integrated semantic retrieval with **Top-k** search and re-ranking over vector databases enabling accurate identification of relevant document segments and improving answer precision compared to traditional keyword-based approaches.
- Designed and implemented **Agentic AI** workflows using **LangGraph**, enabling multi step reasoning, dynamic decision making, and tool invocation (retrieval, summarization, compliance checks) to autonomously generate context aware responses for enterprise use cases.
- Optimized system performance by tuning embedding generation, vector indexing, and prompt strategies, improving response consistency and first response correctness by **25%** through iterative evaluation.
- Built secure and scalable RESTful APIs using **FastAPI**, deployed on **Amazon EC2/EKS**, and exposed via **API Gateway** to support real time query handling and seamless integration with enterprise systems.
- Designed and implemented evaluation and feedback pipelines using curated financial queries to benchmark retrieval and generation quality, enabling continuous improvement in system performance.
- Implemented **query caching** and embedding reuse strategies to reduce redundant computations and improve system efficiency, optimizing performance under repeated query workloads.
- Integrated monitoring and logging pipelines using **CloudWatch** to track performance, query latency, and retrieval accuracy, enabling proactive debugging and ensuring high availability of **GenAI** workflows.

Technical Stack: *AWS Bedrock, LangChain, LangGraph, RAG, Amazon OpenSearch, FAISS, Amazon S3, AWS Lambda, API Gateway, AWS IAM, Amazon CloudWatch, Python (FastAPI, Pydantic), Docker, Amazon EKS, Amazon EC2, AWS Application Load Balancer, AWS VPC, AWS CloudFormation, Boto3, Redis.*

Target | Brooklyn Park, MN
Machine Learning Engineer

Aug 2022 – Jul 2023

Overview: Target's retail operations depend on moving billions of units of inventory across 2,000 stores with surgical precision. This project focused on building a **Centralized Machine Learning platform** to replace legacy statistical forecasting. By integrating real time data such as local weather, social trends, and live clickstream activity, the system predicts SKU level demand with high accuracy.

- Designed a centralized ML platform for demand forecasting and personalization using **Python, PyTorch, TensorFlow and XGBoost**, replacing legacy statistical models and improving prediction consistency.
- Built deep learning recommendation models using **PyTorch and TensorFlow**, improving **CTR by 15%** through customer segmentation, embedding techniques and behavioral feature engineering.
- Architected end-to-end **MLOps pipelines** on GCP leveraging **Vertex AI, Kubeflow and MLflow**, automating data ingestion, model training, validation, and deployment workflows across environments.
- Developed scalable model serving APIs using **FastAPI, Docker, and Kubernetes (GKE)**, handling millions of requests with **sub-100ms** latency during peak events like Black Friday.
- Processed multi terabyte datasets using **PySpark** and BigQuery to build real-time and batch feature pipelines with Redis, reducing feature engineering time by **40%** and enabling low-latency inference.
- Implemented model monitoring with data validation and drift detection pipelines, triggering automated retraining workflows to maintain prediction accuracy in production systems.
- Built distributed training workflows on **Vertex AI** using GPU acceleration, reducing model training time by **30%** and improving experimentation speed for forecasting models.
- Established **CI/CD** and automated infrastructure using **GitHub Actions and Terraform** with GKE auto scaling and multi model endpoints, reducing operational costs by **25%** while maintaining high availability.

- Designed **A/B testing frameworks** to evaluate model variants in production, using statistical analysis and controlled experiments to guide feature rollouts and model selection.
- Collaborated with cross functional teams including Product and Supply Chain to translate business requirements into ML objectives, aligning model outputs with inventory and demand goals.

Technical Stack: *Python, PyTorch, TensorFlow, XGBoost, GCP (Vertex AI, BigQuery, GKE), KubeFlow, MLflow, PySpark, Redis, FastAPI, Docker, Kubernetes, Terraform, GitHub Actions.*

Uber | United States

May 2019 – Nov 2021

Senior Data Engineer

Overview: Developed a real time data platform at Uber to process high volume ride demand, driver availability, and geolocation signals for dynamic pricing and dispatch optimization. Built scalable streaming and batch pipelines using **Kafka and Spark** to compute demand-supply imbalances and generate reliable forecasting features. Integrated these pipelines with downstream pricing and **ML systems** to enable accurate surge pricing and efficient ride allocation. This significantly improved system responsiveness, reduced rider wait times, and enhanced overall platform efficiency under peak traffic conditions.

- Engineered a real time data pipeline using **Apache Kafka and Spark Streaming** to process millions of ride events, enabling dynamic demand supply estimation across geo regions with sub minute latency.
- Designed scalable data models on **Amazon S3, Apache Hive, and Presto**, optimizing partitioning and query performance to support high throughput analytical workloads for pricing and dispatch systems.
- Built batch/streaming feature pipelines via **PySpark** and **Apache Airflow**, integrating them with **Michelangelo** (Uber's ML platform) to deliver curated datasets and improve real time pricing accuracy.
- Optimized end to end pipeline performance and data quality by tuning **Spark jobs** and **Kafka** consumers while implementing rigorous validation checks to ensure reliable inputs for critical decision systems.

Technical Stack: *Apache Kafka, Apache Spark (PySpark, Spark Streaming), Amazon S3, Apache Hive, Presto, Apache Airflow, Michelangelo, SQL, Python, Hadoop Ecosystem.*

eBay | Bengaluru, India

Mar 2017 – Apr 2019

Python Developer

Overview: eBay's global marketplace scalability required high throughput systems to process millions of cross-border merchant listings simultaneously. This project focused on building an intelligent, automated inventory ingestion and validation pipeline within the seller ecosystem. By replacing legacy synchronous checks with distributed processing workflows, the platform enforced strict pricing, localization, and compliance guardrails while significantly reducing bulk upload latency.

- Designed and developed an enterprise scale bulk inventory ingestion platform using **Python (Flask/Django)**, enabling global merchants to sync listings and reducing time to marketplace by **45%**.
- Architected an asynchronous pipeline leveraging **Celery, Redis and RabbitMQ** to validate **15M+** daily listing updates, executing real-time policy checks, currency conversions and metadata sanitization.
- Engineered high performance **RESTful APIs** to seamlessly bridge merchant management tools with internal catalog databases and order fulfillment services, sustaining sub-100ms response times.
- Reduced database read latency by **35%** via **PostgreSQL** query optimization, sharding refinements, and **Memcached**, while maintaining **90%+** test coverage using **PyTest**.

Technical Stack: *Python (Django/Flask), Celery, Redis, MySQL, RabbitMQ, Pillow, PyTest, Docker, Git.*

Certifications

- AWS Certified Generative AI Developer - Professional
- AWS Certified Machine Learning Engineer - Associate
- Microsoft Certified: Azure AI Apps and Agents Developer Associate
- Generative AI Fundamentals - Databricks

Education

University of Maryland, Baltimore County

Master of Science in Computer Science